Questions

Q1.

Carbon-14 is a radioactive substance that emits beta minus (β^-) particles.

(i) Which of these is the unit for the activity of a sample of carbon-14?

(1)

- ☐ **A** Pa
- ☐ **B** Nm
- ☐ **C** Gs
- ☐ **D** Bq

(ii) Which row of the table gives a correct comparison of the penetrating abilities of alpha, beta and gamma radiation?

(1)

	least penetrating ⟶ most penetrating
☐ **A**	alpha, beta, gamma
☐ **B**	alpha, gamma, beta
☐ **C**	gamma, beta, alpha
☐ **D**	gamma, alpha, beta

(iii) The mass of a beta minus particle is 9.1×10^{-31} kg.

Calculate the kinetic energy of a beta minus particle moving with a speed of 3.8×10^{6} m/s.

(2)

kinetic energy = ... J

(Total for question = 4 marks)

Q2.

(a) Positrons are antiparticles of electrons.
The photograph shows a PET scanner that uses positrons to locate the site of a brain tumour.

Positrons are emitted by the radioactive isotope fluorine-18.
The fluorine-18 is attached to glucose.
The glucose is injected into the patient's blood stream and collects at the site of the tumour.

(i) Fluorine-18 has a half-life of 1.8 hours.
State one advantage and one disadvantage of using a substance with such a short half-life.

1. Advantage

(1)

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2. Disadvantage

(1)

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(ii) When a positron meets an electron they annihilate each other.
Explain how this enables the site of the tumour to be located.

(4)

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*(b) Positrons (β+ particles) are emitted from the nuclei of some atoms.
Electrons (β− particles) are emitted from the nuclei of other atoms.
The table gives some information about quarks.

quark	charge (compared to the charge on a proton)
u	+2/3
d	−1/3

Describe, in terms of quarks, how β+ particles are emitted from the nuclei of some atoms and β− particles are emitted from the nuclei of others.

(6)

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(Total for Question = 12 marks)

Q3.

Some rocks contained uranium when they were formed.

Radioactive decay in these rocks produces radon gas.

Explain why people living near these rocks have an increased health risk from background radiation.

(3)

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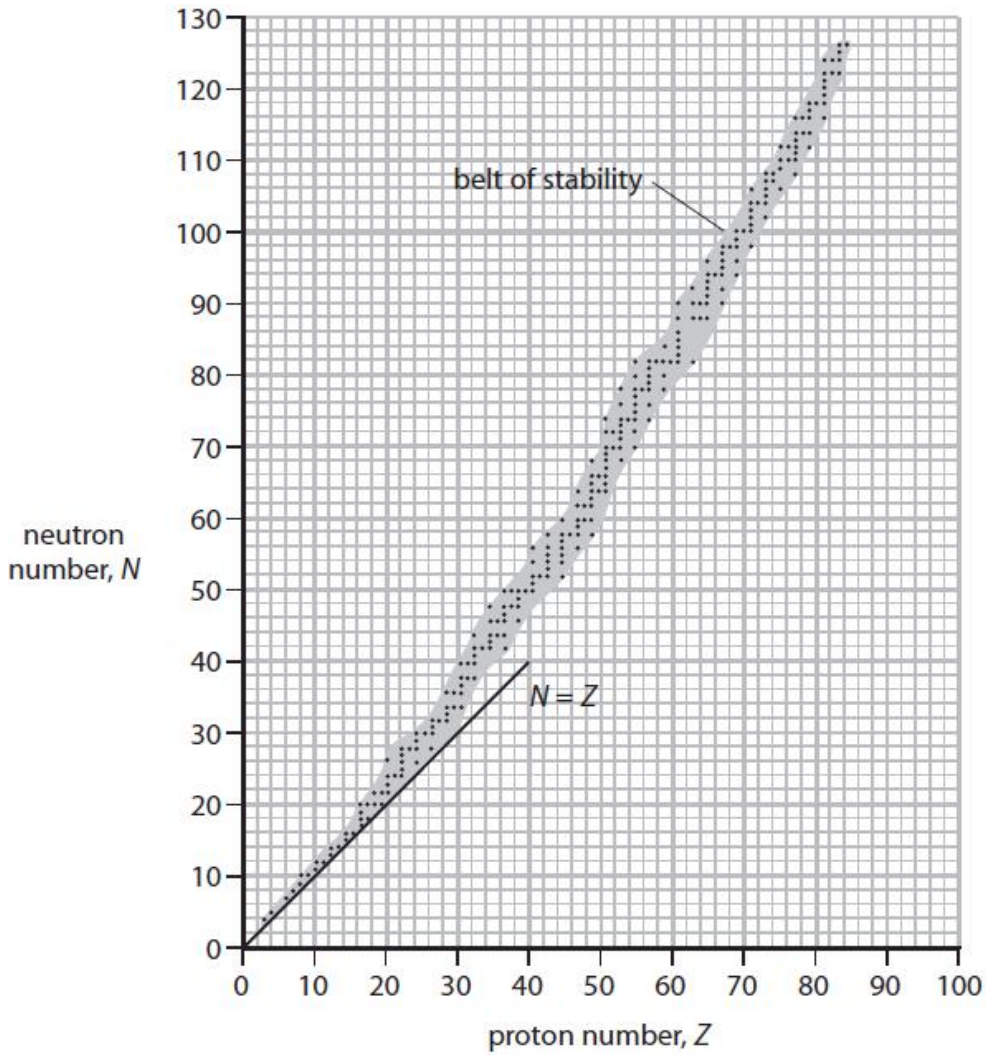
Q4.

(a) The diagram shows the stability curve for nuclear isotopes.



Complete the sentence by putting a cross (☒) in the box next to your answer.
An isotope above the curve will undergo β^- decay because it has

(1)

☐ **A** too few protons
☐ **B** too many protons
☐ **C** too few neutrons
☐ **D** too many neutrons

(b) Which statement is correct for β^+ and β^- particles?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** a β^+ is positively charged and a β^- is negatively charged
☐ **B** the mass of a β^+ is 1800 times the mass of a β^-
☐ **C** the charge on a β^+ is twice the charge on a β^-
☐ **D** a β^+ is a proton and a β^- is an electron

(c) Carbon-14 decays by emitting a β^- particle to form an isotope of nitrogen.

Complete the nuclear equation for this decay by filling in the boxes.

(2)

$$^{14}_{6}C \rightarrow \begin{array}{c}\square\\ \square\end{array}N + \begin{array}{c}\square\\ \square\end{array}\beta^{-}$$

(d) Protons and neutrons both contain quarks.

Describe the arrangement of quarks in a proton.

(2)

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(e) Explain what happens to a nucleus during β^{+} decay.

(2)

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Q5.

A G-M tube is connected to a counter.
A teacher places the G-M tube near to a radioactive source.
A student starts the counter and clock at the same time and writes down the readings shown on the counter every 15 s.

The student plots the readings with a line of best fit, as shown in Figure 10.

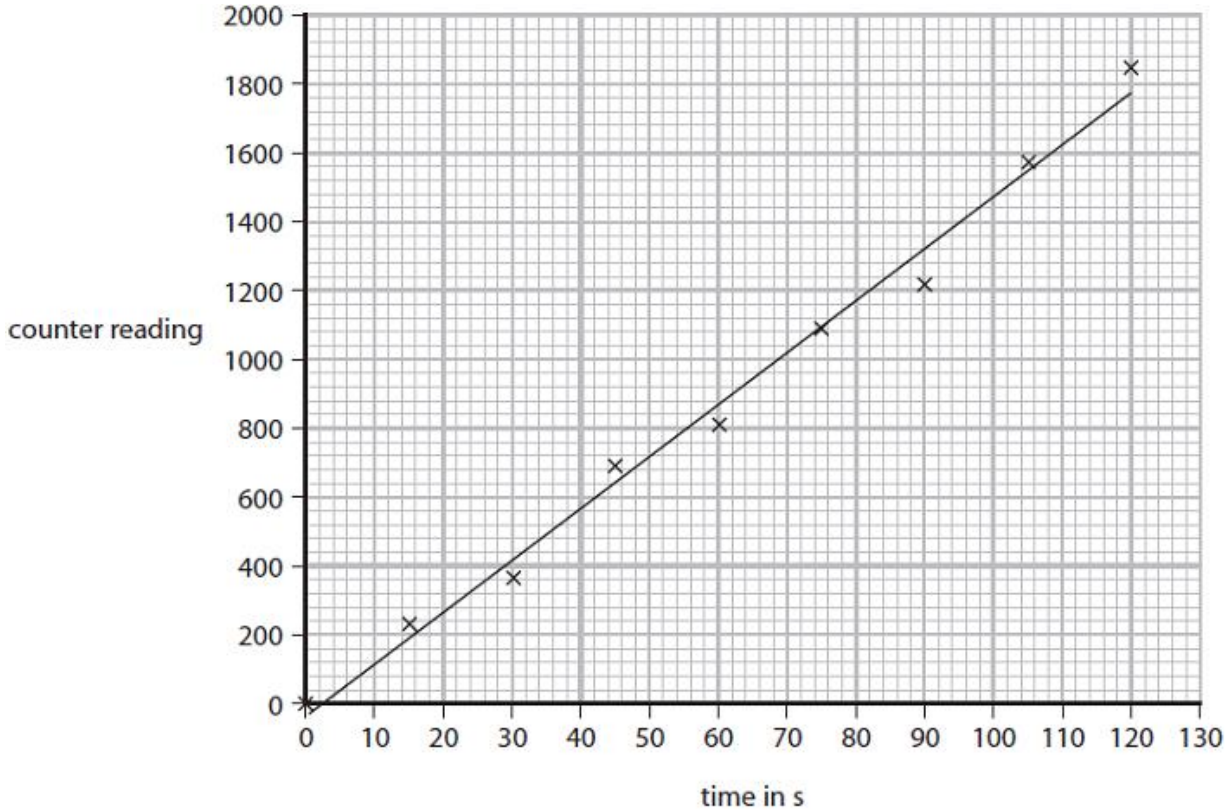

Figure 10

(i) Calculate the average count rate, in counts / s, from the graph.

Show your working on the graph.

(2)

average count rate = ... counts / s

(ii) The student says that the experiment must have been done carelessly because the data seemed quite scattered
away from the best fit line.

The teacher claims such results should be expected in radioactivity experiments.
Justify the teacher's claim.

(2)

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(Total for question = 4 marks)

Q6.

This question is about radioactivity.

Radium-223 is a radioactive substance.

Radium-223 is an alpha emitter.

The half-life of radium-223 is 11 days.

A radioactive source contains 1.7×10^{23} nuclei of radium-223.

Calculate the number of radium-223 nuclei remaining in the source after a time of 33 days.

(2)

number of radium-223 nuclei remaining = ...

(Total for question = 2 marks)

Q7.

The isotope fluorine-18 is used in PET scans, which help to detect cancers.

Figure 7 shows a radioactive decay curve for a sample of fluorine-18.

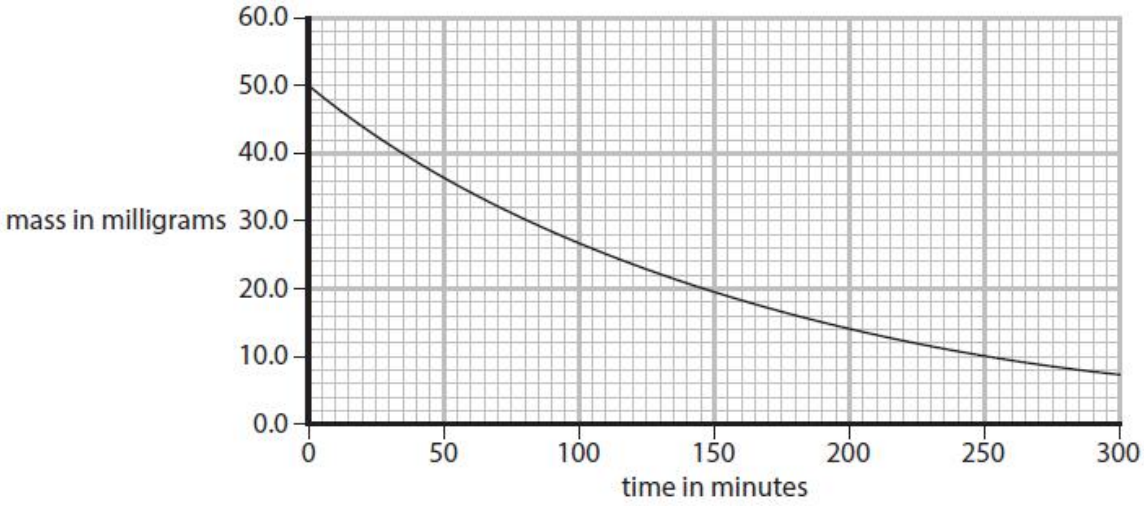

Figure 7

(i) Use the graph to calculate the percentage of fluorine-18 remaining after 150 minutes.

(2)

percentage of fluorine-18 remaining after 150 minutes = ... %

(ii) Fluorine-18 releases positrons.

Describe what happens when positrons meet electrons.

(2)

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(Total for question = 4 marks)

Q8.

In the Sun, four protons start the process of nuclear fusion.

These protons combine and finally produce a helium nucleus.

The helium nucleus has a smaller mass than the four protons.

This difference in mass is converted to energy.

Four protons have a total mass of 6.69×10^{-27} kg.

A helium nucleus has a mass of 6.64×10^{-27} kg.

Calculate the percentage of the original mass that has been converted to energy.

(3)

percentage of mass converted to energy ..... %

(Total for question = 3 marks)

Q9.

In 2023, Finland produced 4.4 GW of power from nuclear power plants.

The UK produced 8.9 GW of power from nuclear power plants.

Calculate the ratio of nuclear power produced in Finland compared with the UK as

$$\frac{\text{power production in Finland}}{\text{power production in the UK}}$$

Give your answer to 2 significant figures (2 s.f.).

(2)

ratio = ...

(Total for question = 2 marks)

Q10.

The intensity of gamma radiation can be measured using a Geiger-Müller tube and counter.

The count rate recorded by the counter tube depends on how far away the Geiger-Müller tube is from the gamma radiation source.

The equation relating count rate to distance from the source is

$$\text{count rate} = \frac{k}{d^2}$$

where d is the distance from the source and k is a constant.

A Geiger-Müller tube is placed 0.70 m from a source of gamma radiation.
The counter displays a count rate of 85 000 count per minute.

Calculate the count rate recorded when the Geiger-Müller tube is placed 1.3 m away from the same gamma radiation source.

(3)

count rate = ... counts per minute

(Total for question = 3 marks)

Q11.

Radioactive tracers can be used when scanning a person's kidneys.

A radioactive isotope is injected into a person's blood stream.

The isotope emits radiation.

As the blood flows through the kidneys, this radiation is detected outside the body by a scanner.

(i) What type of radiation travels from the kidney to the scanner?

(1)

- ☐ **A** alpha
- ☐ **B** beta plus
- ☐ **C** beta minus
- ☐ **D** gamma

(ii) During the scan, a technician needs to take readings for about 30 minutes.

The half-life of the isotope used is about 6 hours.

1. State why an isotope with a half-life of about 6 minutes is **not** suitable.

(1)

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2. State why an isotope with a half-life of about 6 days is **not** suitable.

(1)

.....

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(iii) State **two** ways of reducing the radiation risks to the technician.

(2)

1 .....

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2 .....

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(Total for question = 5 marks)

Mark Scheme

Q1.

Question Number	Answer	Mark
(i)	**The only correct answer is D Bq** ***A*** *is not correct because it is the unit of pressure* ***B*** *is not correct because it is the unit of moment* ***C*** *is not correct because it is the unit of time*	**(1)** **AO1.1**

Question Number	Answer	Mark
(ii)	**The only correct answer is A** ***C and D*** *are not correct because they start with most penetrating* ***B*** *is not correct because gamma is more penetrating than beta*	**(1)** **AO1.1**

Question Number	Answer	Additional guidance	Mark
(iii)	selection of (KE =) ½ mv^2 (1) evaluation (1) 6.6×10^{-18} (J)	$\frac{1}{2} \times 9.1 \times 10^{-31} \times [3.8 \times 10^{6}]^2$ allow values that round to 6.6×10^{-18} (J) e.g. 6.5702×10^{-18} (J) award full marks for the correct answer without working	**(2)** **AO2.1**

Q2.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	1. advantage (1) minimises patient's exposure to radioactivity 2. disadvantage (1) (has to be produced) close to the {place / time} of use	Does not stay in the (patient's) body for a long time / Decays quickly Any time constraint (in diagnosis/scanning/treatment) Ignore confusion between biological and physical half-life	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	An explanation linking four from: • gamma (rays emitted) (1) • two (gamma rays) (1) • in opposite directions (1) • (because) momentum is conserved (1) • detectors / sensors placed around the patient (1) • simultaneous detection (1)	pair at 180°(to each other) gamma cameras/scintillation crystals idea of triangulation	**(4)**

Question Number		Indicative Content	Mark
QWC	*** (b)**	An explanation including some of the following points • protons and neutrons are made up of quarks • quarks can change (flavour) • proton made up of uud • neutron made up of udd • for β^+ a u changes to a d • for β^- a d changes to a u • for β^+ a p changes to a n • for β^- a n changes to a p	**(6)**
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited description e.g. for β^+ a proton changes to a neutron OR protons and neutrons are made up of quarks which can change (flavour) • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple description including both processes but with significant error(s) e.g. 'a d changes to a u' [error], changing a proton to a neutron and emitting a β^+. A u changes to a d [error] changing a neutron to a proton and a β^- [errors in u and d] OR one of the processes with the correct detail e.g. A u goes to a d therefore a proton goes to a neutron and emits a β^+ • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed, accurate description of both processes e.g. A u goes to a d therefore a proton goes to a neutron and emits a β^+. A d goes to a u therefore a neutron goes to a proton and emits a β^-. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

(Total marks for question = 12 marks)

Q3.

Question Number	Answer	Acceptable answers	Mark
	An explanation linking any three from • Radon is radioactive (1) • Radon can escape from rocks and buildings (1) • Radon can be inhaled (1) • Radiation (from radon) can cause cancer (1) • Radon emits alpha (1)	Ignore radiation from rocks themselves Radon enters/gets trapped in buildings/homes / increases background radiation (breathed into) lungs (DNA) mutation / cell damage (Highly) ionising radiation	**(3)**

Q4.

Question Number	Answer	Acceptable answers	Mark
(a)	☒ **D** too many neutrons.		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	☒ **A** a β^+ is positively charged and a β^- is negatively charged		**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	$^{14}_{7}N + ^{0}_{-1}\beta^-$ Any two numbers correct (1) All four numbers correct (2)		**(2)**

Question Number	Answer	Acceptable answers	Mark
(d)	A description to include: Up and down (quarks) / Three (quarks) (1) Correct arrangement (quarks) (1)	 Accept for two marks: uud up, up, down two up quarks and one down quark Ignore charges	**(2)**

Question Number	Answer	Acceptable answers	Mark
(e)	An explanation linking the following: **Either** proton changes to a neutron (1) positron/anti-electron (emitted) (1) OR up quark changes to a down quark (1) positron/anti-electron (emitted) (1) **OR** proton number goes down by one / neutron number goes up by one (1) number of nucleons stays the same (1)	Accept any correct set of statements for two marks $P \rightarrow n + \beta^{+}$ (1) Ignore positive electron atomic number goes down by one mass number is constant	**(2)**

Q5.

Question number	Answer	Additional guidance	Mark
(i)	use of gradient on graph (1) $= \frac{1480}{97}$ evaluation (1) 15.3 (counts /s)	look for a triangle / line going up allow $\frac{1480}{100}$ accept other data from the graph allow numbers between 12.0 and 16.0 award full marks for answers in the correct range without working	(2)

Question number	Answer	Additional guidance	Mark
(ii)	explanation the process (of radioactive decay) is unpredictable / (occurs) random(ly) (1) so the count rate would not be constant / there will be variations with each reading (1)	do not allow 'difficult to predict' ignore background results (expected to) scatter	(2)

Q6.

Question number	Answer	Additional guidance	Mark
CS2	33 days is 3 half-lives (1) 2.1(25) × 10^{22} (1)	$\frac{1.7 \times 10^{23}}{2}$ (× 2 × 2) 2.1(25) to any other power of ten scores mp1 only award full marks for correct answer without working.	**(2)** **AO2**

Q7.

Question number	Answer	Additional guidance	Mark
(i)	reading from graph and substitution (1) = $\frac{19.5}{50}$ (x100) evaluation (1) 39 (%)	allow use of values between 19 and 20 inclusive allow values between 38(%) and 40(%) inclusive award full marks for correct answer without working.	**(2)** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	a description to include any **two** from they collide (1) annihilate (1) (releasing a pair of) gamma ray(s) (1)	 ignore interact / combine / attract /meet allow (charges) cancel / neutralise / become neutral photons release energy ignore light	**(2)** **AO1.1**

Q8.

Question	Answer	Additional guidance	Mark
	subtraction (1) $6.69 \times 10^{-27} - 6.64 \times 10^{-27}$ OR $0.05 (\times 10^{-27})$ OR $5.0 (\times 10^{-29})$ use of % (1) $\frac{5.0 \times 10^{(-29)} \times 100}{6.69 \times 10^{-27}}$ evaluation (1) 0.75 %	accept negative mass difference throughout accept use of 0.05×10^{-27} 6.64×10^{-27} 0.75/0.747/0.753 to any other power of 10 scores 2 marks $\frac{6.64 \times 10^{-27}}{6.69 \times 10^{-27}} (\times 100\%)$ or 99% or 0.99 scores 1 mark award full marks for the correct answer without working	**3** **AO2.1**

Q9.

Question number	Answer	Additional guidance	Mark
	substitution (1) $= \frac{4.4\ (\times 10^9)}{8.9\ (\times 10^9)}$ evaluation to 2 s.f.(1) 0.49	 0.49438 allow 4.4 : 8.9 allow 44:89 allow 0.49 : 1(.0) accept 1(.0) : 0.49 award full marks for correct answer without working.	**(2)** **AO2.1**

Q10.

Question number	Answer	Additional guidance	Mark
	Substitution and rearrangement to find k (1) $k = 85000 \times 0.70^2$ Substitution to find new count rate (1) count rate $= \frac{85000 \times 0.70^2}{1.3^2}$ Answer (1) 25000 (counts per minute)	 41650 24645 (counts per minute)	 **(3)**

Q11.

Question Number	Answer	Mark
(i)	**D gamma** ***A*** *is not correct because alpha cannot pass through and out of the body* ***B*** *is not correct because beta plus cannot pass through and out of the body* ***C*** *is not correct because beta minus cannot pass through and out of the body*	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii) **CLIP with** **(ii)**	decays too quickly to give a reading (1)	accept (half-life) not long enough for reading to be taken ignore disappear after 12 min	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii) **CLIP with** **(ii)**	stays in the body too long (1)	accept could harm / damage other organs patients stay radioactive for too long so the patient does not get too high a dose of radiation	**(1)** **AO1**

	Answer	Additional guidance	Mark
(iii)	**Two** from: shielding (1) time limiting exposure (1) distance limiting exposure (1) wear PPE / protective clothing (1)	 accept stand behind barriers / store (source) in lead box radiation monitoring badges leave the room / go outside / stay away from the patient / use tongs lead aprons / gloves ignore goggles / masks	**(2)** **AO1**